graphs, you will find GeoGebra very effective. It is a powerful tool to teach interactive geometry, calculus, statistics and algebra. It is designed to incorporate the needs of learners, starting from schools to higher education.

GeoGebra is available as a software application, specific to platforms such as Windows and Linux *(https://www.geogebra.org/)*. It is also available as an app for popular platforms such as Android and iOS. The major features of GeoGebra are listed below:

- Interactive geometry environment
- Spreadsheet availability
- Inbuilt statistics and calculus tools
- Ability to script
- Availability of pre-built learning resources through GeoGebra materials (*https://www.geogebra.org/materials*)

Detailed documentation is available in the official website. Various links to books are also provided (*https://www.geogebra.org/m/XUv5mXTm*). Though GeoGebra is not an integral part of all e-learning environments, it could be very effective for those teaching subjects such as mathematics and statistics.

Elgg for building a social network

Elgg is a powerful social networking engine that is open source. This robust framework facilitates the building of a campus-wide social network for colleges or universities.

Elgg can be downloaded from the official website *https://elgg.org/*. Detailed installation instructions are available at *http://learn.elgg.org/en/stable/intro/install.html.*

Successful use of Elgg is showcased at *https://elgg.org/showcase.*

Tools such as Elgg can be used to make the communication process effective in the e-learning environment.

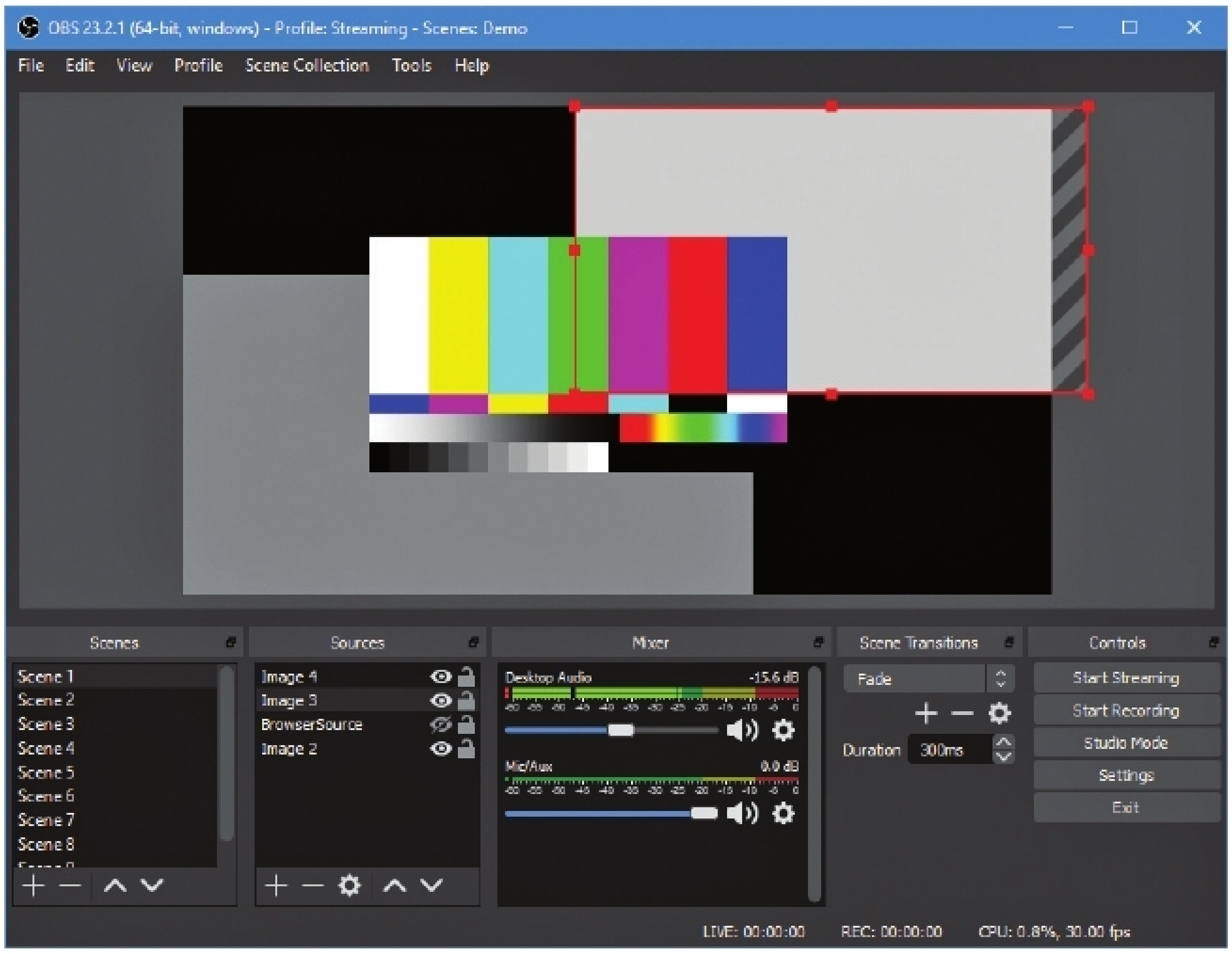

Figure 6: OBS Studio

OBS Studio for screen recording

Desktop screens can be captured to build e-learning video content. There is a lot of software to assist in this process, but I would suggest the use of OBS (Open Broadcaster Software) Studio for the purpose of screencasting and live streaming. The major features of OBS are:

- Free and open source
- Available across all major platforms
- High performance real-time video/audio capturing
- The capability to set up an unlimited number of scenes and to switch between them, seamlessly
- Easy-to-use audio mixer
- Modular Dock UI, which enables repositioning of elements as per your convenience
- Support for all major streaming platforms
- Availability of a powerful API

A screenshot from the official website is shown in Figure 6 (*https://obsproject.com/*).

Screencasting and streaming can be carried out in a professional manner if you familiarise yourself with the features of OBS Studio.

There are plenty of other such open source tools available to assist in e-learning. This article is just a starting point to help you explore tools that match your requirements. END

References

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